Kartik Mangal

Software Engineer II | Gameplay Programmer | LiveOps, Multiplayer & Gameplay Systems

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SUMMARY

Gameplay programmer with 6 years of experience in Unity and C# across LiveOps, multiplayer, real-money casino and mobile games. Builds new features, add-on mechanics and A/B-tested experiments in Lua and C# for a live mobile game, with a background in Photon Fusion and SignalR multiplayer, physics-based gameplay and Unity Editor tooling.

EXPERIENCE

Software Engineer II, Liquidnitro Games

Jun 2025 - Present

Unity, C#, Lua, LuaPerfect, Excel configs, Photon Fusion, VS Code, Claude Code

LiveOps Mobile Title | Feb 2026 - Present

- Joined the team on day one of a five-year-old live IP moving to Liquidnitro and helped ship 15 patch releases since Feb 2026.
- Built new features and add-on mechanics for live players in Lua on top of Unity and C#, debugging with LuaPerfect attached to the running Unity project.
- Ran A/B-tested gameplay experiments end to end, contributing to an increase in ARPU.
- Turned Excel design configs, the game's main entry point for tuning and content, into shippable features.
- Ramped up on a large live codebase by debugging it first-hand, then used Claude Code with a token-efficient workflow to deliver faster without dropping quality.

Feature Development and Bug Fixing | Nov 2025 - Feb 2026

- Implemented gameplay features end to end on an unreleased title, from design spec to a tested build.
- Investigated, reproduced and fixed bugs across the project to keep builds stable.

Multiplayer Mini Games | Jun 2025 - Nov 2025

- Designed and implemented network-synced gameplay mechanics using Photon Fusion, keeping dynamic player and environment interactions in sync across all clients.
- Developed a dynamic path generation system and an obstacle hit-detection and avoidance system that adapts in real time to gameplay, for both player and AI navigation.
- Worked closely with designers and artists to optimise the performance and latency of multiplayer interactions.

Game Developer, Arrise Solutions Pvt. Ltd.

Jun 2023 - May 2025

Unity 3D, C#, SVN, Jenkins, Spine, NGUI, UGUI

Bespoke Slot Games

- Developed three categories of real-money slot games for web and mobile.
- Integrated art, multi-lingual fonts and animation using NGUI and the Spine controller.
- Carried out QA testing, debugging and bug-fixing.
- Optimised assets for performance and compatibility across a wide range of devices.

Unity Developer, BR Softech

Mar 2023 - May 2023

Unity 3D, C#, SignalR, GitHub Desktop

Crash Casino Game Re-skin

- Delivered full-stack development on a re-skin of a crash-style casino game for mobile, working across the Unity client and the C# backend.
- Integrated SignalR real-time networking between the Unity client and the server for live game rounds.
- Applied the new theme, art and UI across the mobile client.

Game Developer, Totality Corp — Oct 2020 - Dec 2022

Unity 3D, C#, Photon PUN, Roblox, Lua

Vijayi Dash | Oct 2022 - Dec 2022

- Developed a complete obstacle system using ScriptableObjects and DOTween.
- Used the State design pattern and the New Input System to handle player locomotion.
- Used the Observer pattern to avoid runtime exceptions and unnecessary per-tick memory allocations.
- Implemented physics-based collision detection with primitive colliders and Rigidbody components.
- Optimised performance using profiling-led techniques.

Rann Bhumi | Nov 2021 - Sep 2022

- Developed a deathmatch combat game with five character classes, each with its own weapons and abilities.
- Built physics-based locomotion, modular weapon systems, UI screens and dynamic camera behaviours.
- Integrated PvP networking, host migration, RPCs for client- and server-specific logic, and game state synchronisation.

PERSONAL PROJECTS

WATERMARK — Sep 2026 - Present

Unity 6, URP 17, C#, Cinemachine, New Input System, ScriptableObjects, Claude Code

- Turned a jelly-sphere character into a body made of water that wobbles, stretches on the run, squashes on landing, sheds droplets and melts into a puddle on a dive.
- Built a Water Launcher whose shots are paid for out of the shooter's own body, landing as a splash (droplet crown, jets, spatter) and a puddle that spreads, dries up and can be stood in to refill.
- Split the system into three layers (gameplay numbers, a light CPU droplet simulation, a GPU-drawn water skin) so the visuals never feed back into gameplay, keeping the shared state small enough to network.

Blockout Studio — Sep 2026 - Present

Unity Editor tooling, C#, UPM package, Python, Preact, Blender, Claude Code

- Building an embeddable Unity package that turns a reference image or text prompt into a dimensioned, playable greybox level with colliders, clip walls and out-of-bounds zones.
- Built a desktop studio (Python and Preact) that drives a running Unity Editor over a local bridge, with Blender sync for hand edits.
- Added a Theme Kit that restyles a whole level from a single reference image.

TECHNICAL SKILLS

Languages: C#, Lua

Engines & Tools: Unity 6 / URP, Unity 3D/2D, Visual Studio, VS Code, LuaPerfect, Blender (level bridge)

LiveOps: Patch release cycles, A/B-tested experiments, Excel-driven design configs, Live debugging

Gameplay: Physics-based locomotion, Cinemachine, New Input System, Spine, DOTween

Networking: Photon Fusion, Photon PUN, Netcode for GameObjects

Architecture & Optimisation: HLD, LLD, SOLID, Design patterns, ScriptableObjects, Batching, Profiler, Debugging & testing

Tooling & AI: Unity Editor tooling, UPM packages, Claude Code (skills, agents, hooks), Unity CLI automation

UI: NGUI, UGUI / Canvas

Workflow: Perforce, Git / GitHub, SVN, Jenkins, JIRA, Notion

EDUCATION

Indian Institute of Information Technology, Jabalpur — Jul 2018 - May 2022

B.Tech, Electronics and Communication Engineering | Jabalpur, India